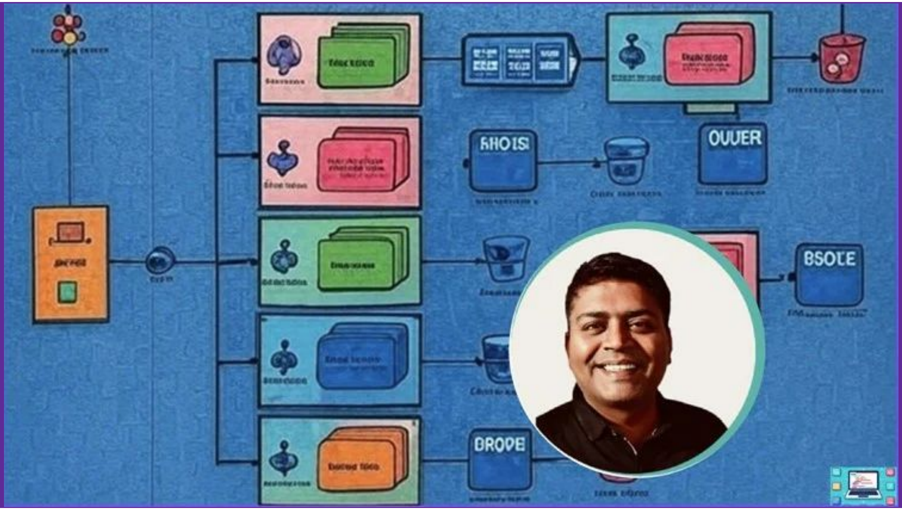
Introduction to Security in System Design

Security in System Design - Principles, Practices & Protocols

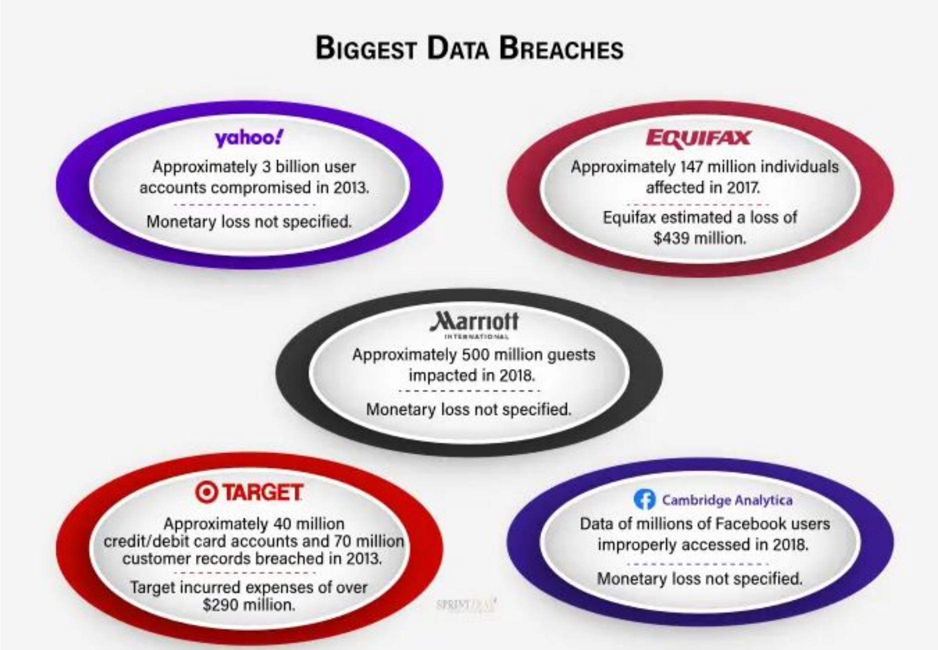
Mastering System Design: From Basics to Cracking Interviews

Rahul Rajat Singh



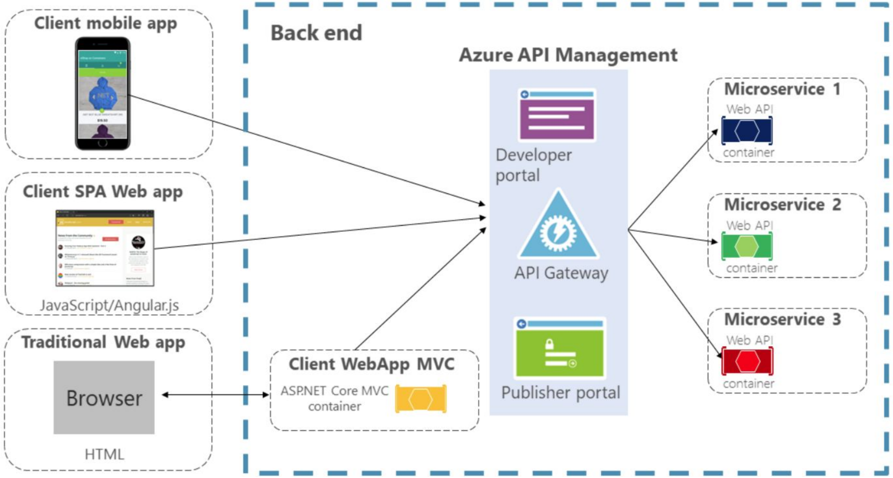
Why Security Matters in System Design

- Security is a non-functional requirement but critical for:
  - User trust
  - Data protection
  - System reliability
- Real-world examples: Data breaches (Equifax, Facebook, etc.)



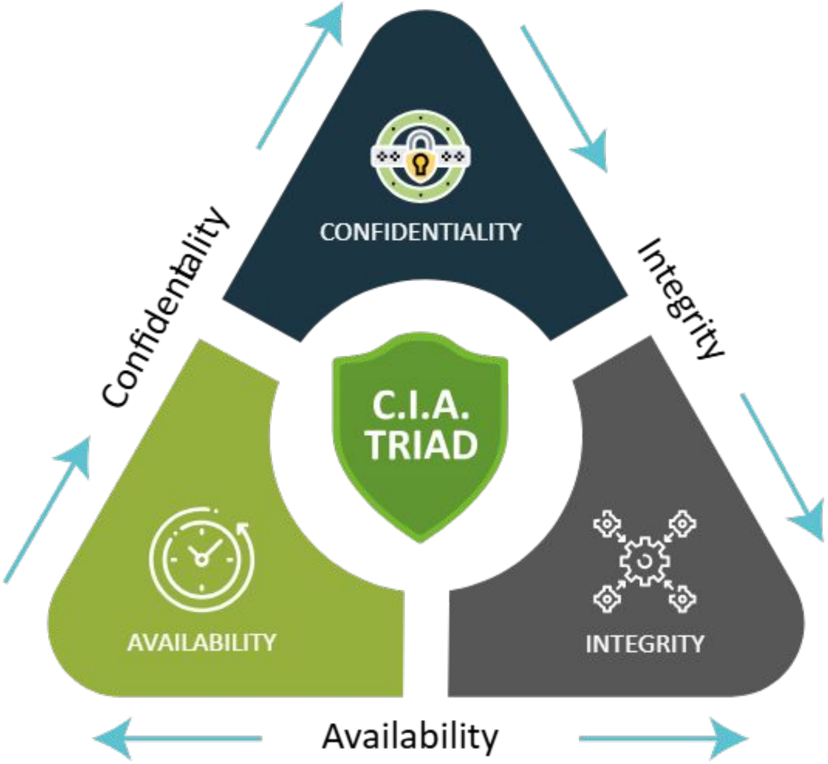
What is Security in Distributed Systems?

- Distributed systems: more entry points, more vulnerabilities
- Security considerations:
  - Data in transit & at rest
  - Authentication, Authorization
  - Secure APIs & endpoints
  - Node and network-level protection



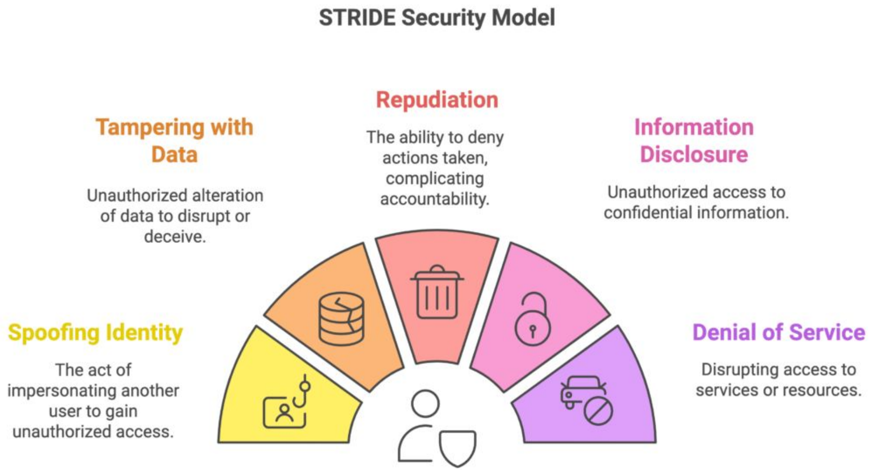
The CIA Triad: Core of System Security

- Confidentiality: Prevent unauthorized access
- Integrity: Prevent data tampering
- Availability: Ensure system uptime and access



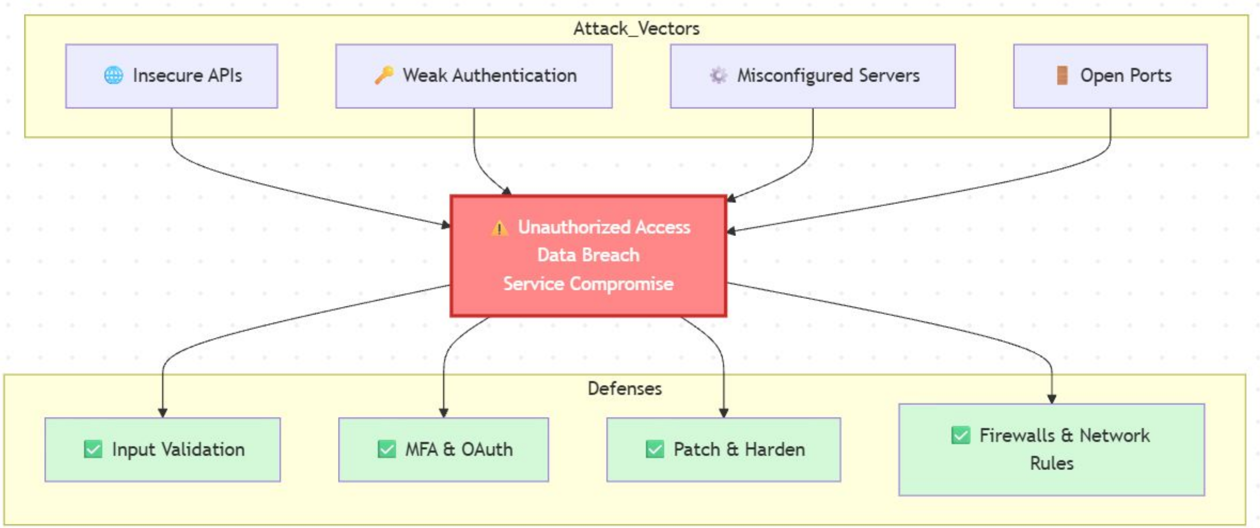
Threat Modeling: Understanding Your Adversary

- Define potential attackers and what they want
- Consider:
  - Attack surface
  - Entry points
  - Assets to protect
- Tools: STRIDE model (Spoofing, Tampering, Repudiation, Info disclosure, Denial of service, Elevation of privilege)



Common Attack Vectors

- How attackers get in:
  - Insecure APIs
  - Misconfigured servers
  - Poor authentication
  - Open ports / services



Common Attacks

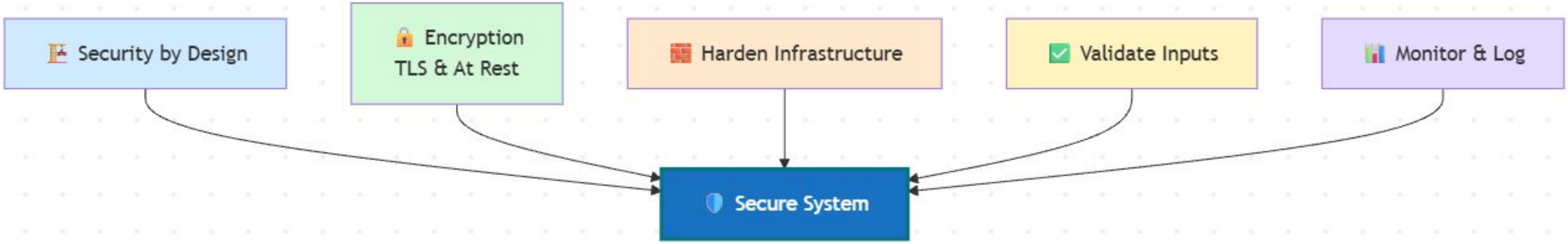
- DDoS (Distributed Denial of Service)
  - Flooding system with traffic to disrupt service
  - Target: Availability
  - Mitigation:
    - Rate limiting
    - Traffic scrubbing (e.g., Cloudflare)
    - Autoscaling and failover strategies
- Man-in-the-Middle (MITM) Attack
  - Attacker intercepts communication
  - Targets: Confidentiality & Integrity
  - Protection:
    - HTTPS (TLS)
    - Certificate pinning
    - VPNs
- Injection Attacks (e.g., SQL Injection)
  - Attacker sends malicious input to execute unwanted commands
  - Impacts: Data integrity, confidentiality
  - Mitigation:
    - Input validation
    - Parameterized queries
    - WAF (Web Application Firewall)
- Spoofing Attacks
  - Impersonation of another user/system
  - Types: IP spoofing, email spoofing, DNS spoofing
  - Defense:
    - Multi-factor authentication
    - Token-based auth
    - IP whitelisting

Security in the Software Development Lifecycle (SDLC)

- Embed security from the start (Shift Left)
- Phases:
  - Requirements: Threat modeling
  - Design: Secure architecture
  - Development: Secure coding
  - Testing: Security tests, fuzzing
  - Deployment: Secrets management
  - Maintenance: Patch management

Best Practices

- Adopt security by design
- Use encryption (TLS, at-rest)
- Harden infrastructure (firewalls, VPCs)
- Validate inputs and sanitize outputs
- Monitor and log activity



Interview Questions – Security Focused

- How would you design a secure authentication system for a distributed application?
- Explain how the CIA triad applies to system design.
- What are common security threats in a microservices architecture, and how would you mitigate them?
- How would you protect your system from a DDoS attack?
- What role does TLS/HTTPS play in system security?
- And how would you implement certificate management at scale?
- How can you ensure secure data storage in a cloud-based system?
- What is threat modeling and how would you incorporate it into your design process?

Summary and Key Takeaways

- Security = ongoing process, not a one-time task
- Understand the CIA triad
- Identify and address threat vectors
- Build secure systems from the ground up
- What’s next:
  - Authentication & Authorization

Authentication & Authorization

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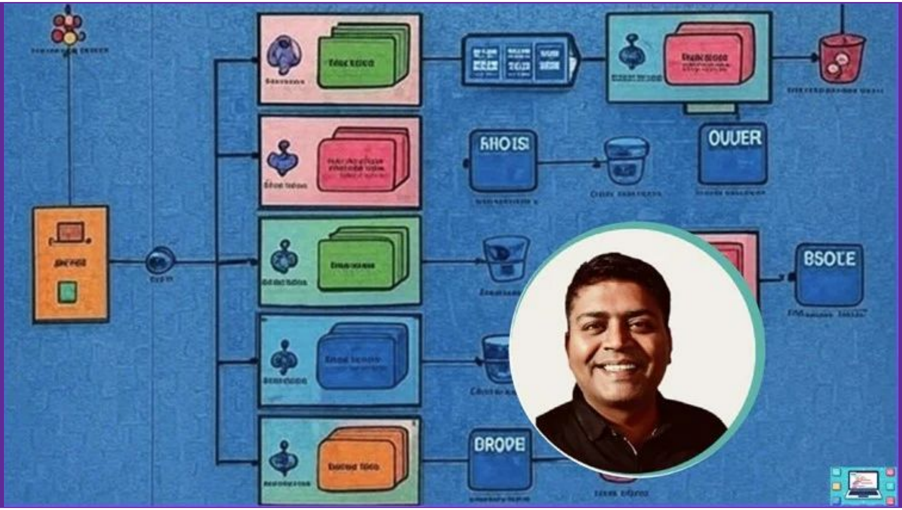
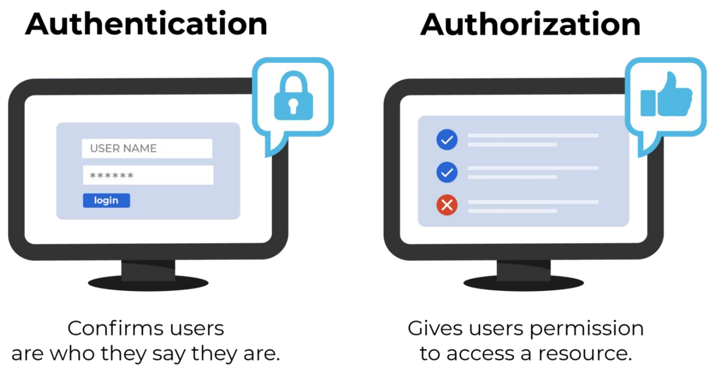


Introduction to Authentication vs. Authorization

- Authentication: Verifying who the user is (identity verification).
- Authorization: Granting the user permission to access specific resources based on their identity.
- Key difference: Authentication is about who you are, while Authorization is about what you can do.

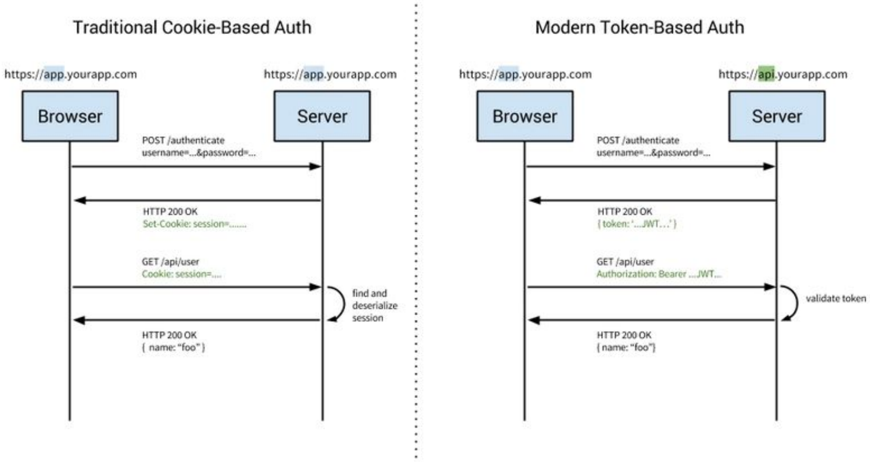


Common Authentication Methods

- Basic Authentication: Simple user and password-based authentication.
- OAuth2: Delegated access protocol, allowing third-party services to access user data without exposing credentials.
- OpenID Connect: An identity layer built on top of OAuth2 for authentication, often used for single sign-on (SSO).
- JWT (JSON Web Tokens): Token-based authentication, commonly used in stateless applications and APIs.

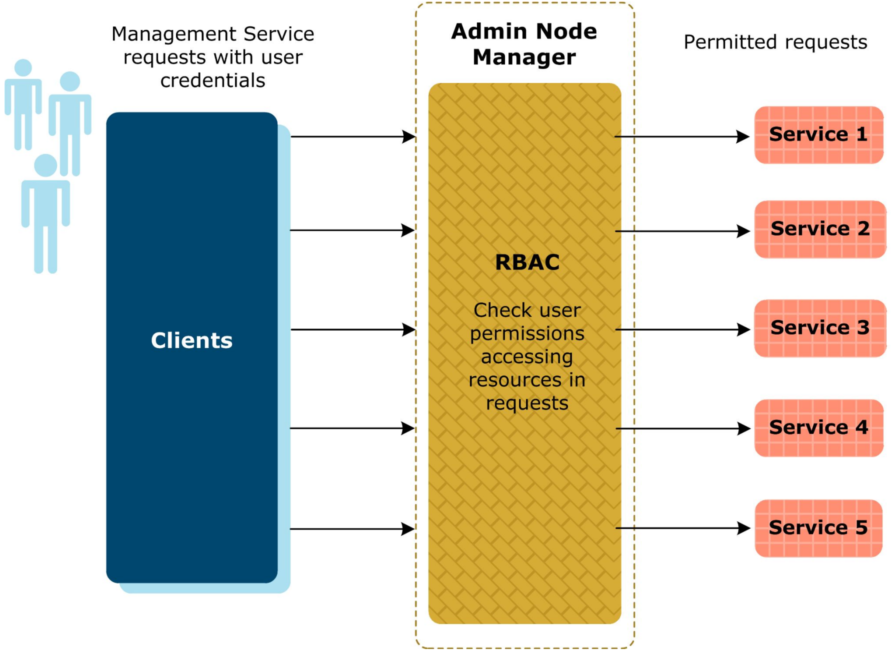
Session-based vs. Token-based Authentication

- Session-based Authentication: Server-side storage of session data, typically using cookies.
  - Pros: Easy to implement, works well with traditional web applications.
  - Cons: Scalability challenges in distributed systems.
- Token-based Authentication: Stateless, where tokens (e.g., JWT) are used to authenticate the user.
  - Pros: Scalable, decouples backend systems.
  - Cons: Requires secure token storage and handling.



Access Control Models

- RBAC (Role-Based Access Control): Assigns permissions based on user roles (e.g., Admin, User, Viewer).
  - Simpler to manage but less flexible.
- ABAC (Attribute-Based Access Control): Grants access based on attributes (e.g., department, project).
  - More fine-grained but more complex.
- DAC (Discretionary Access Control): Owner of a resource defines access controls.
- MAC (Mandatory Access Control): Access control decisions are made by a central authority based on security policies.



SSO (Single Sign-On) and Identity Federation

- SSO: A user authentication process that allows a user to access multiple applications with a single set of credentials.
  - Benefits: User convenience, reduced password fatigue.
- Identity Federation: A system where multiple identity providers (e.g., Google, Facebook) can be used to authenticate users across different platforms.
  - Benefits: Seamless user experience across different services, reduces need for creating multiple accounts.

Interview Questions for Authentication & Authorization

- Why is JWT commonly used for stateless authentication in distributed systems?
- How does OAuth2 differ from OpenID Connect?
- Explain the difference between RBAC and ABAC. Which one is more suitable for a highly dynamic environment?
- What are the advantages of using Single Sign-On (SSO) in a system?
- What are some potential security concerns with token-based authentication?

Summary and Key Takeaways

- Authentication verifies the user's identity, while authorization defines what the user can access.
- Common authentication methods include Basic Auth, OAuth2, OpenID Connect, and JWT.
- Access control models include RBAC, ABAC, DAC, and MAC.
- SSO and identity federation simplify user authentication across multiple platforms.
- What's next:
  - Data Protection & Secure Communication

Data Protection & Secure Communication

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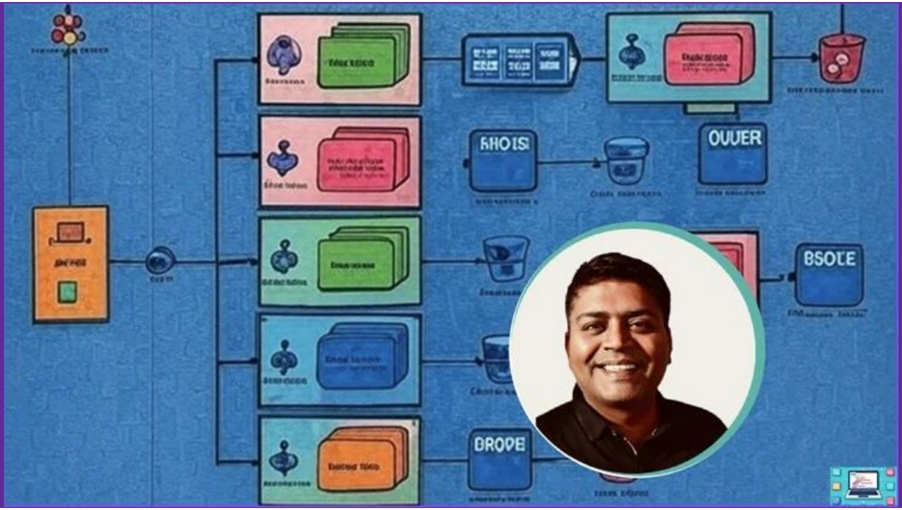


Why Data Protection Matters

- Growing threats: breaches, man-in-the-middle attacks, data leaks
- Regulations (GDPR, HIPAA, etc.)
- User trust and system reliability

Let’s talk Encryption

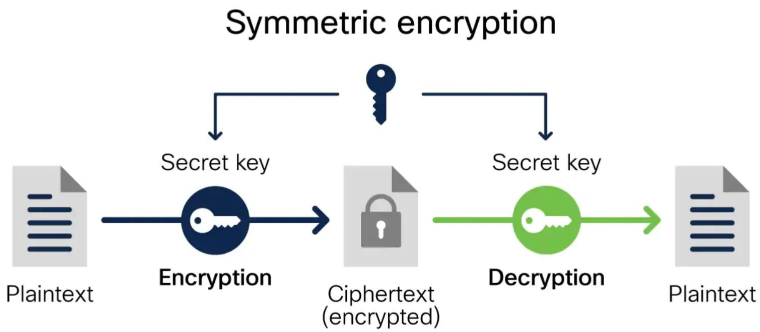
- What is encryption?
- Plaintext → Ciphertext → Decryption
- Key = secret for encoding/decoding

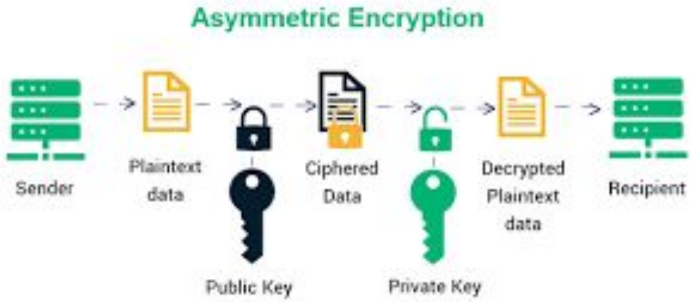
Encryption at Rest and Transit

- Encryption at Rest
  - Protects stored data (disks, databases, backups)
  - Common techniques: Full-disk encryption, database-level encryption
  - Use cases: cloud storage, user files, logs
- Encryption in Transit
  - Secures data during transmission (e.g., HTTP request/response)
  - TLS/SSL protocols enable secure communication
  - Must-have for APIs, user sessions

Symmetric vs. Asymmetric Encryption

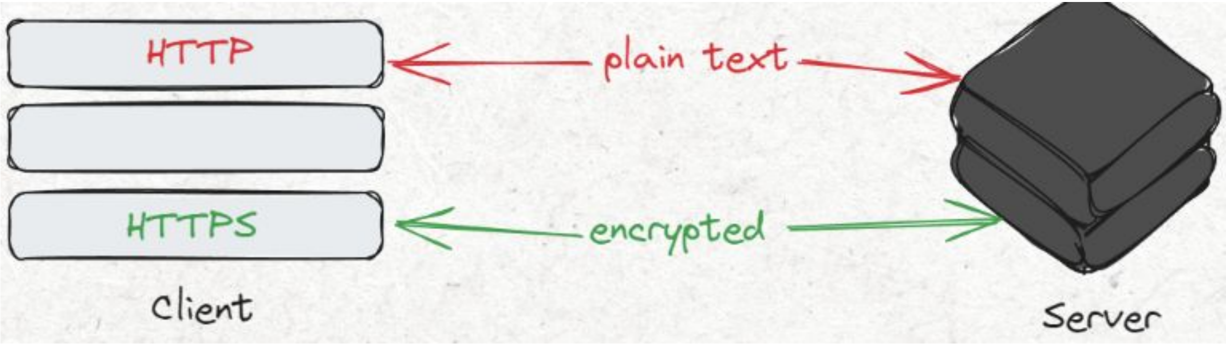
- Symmetric: one key (fast, used for large data)
- Asymmetric: public/private key pair (secure key exchange)
- Often used together (e.g., TLS handshake)





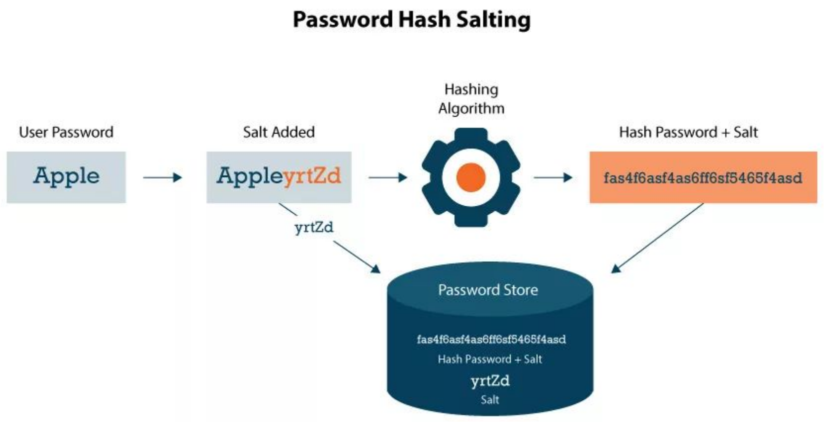
TLS/SSL and HTTPS

- HTTPS = HTTP over TLS
- Ensures confidentiality, integrity, and authenticity
- TLS handshake: key exchange + cipher negotiation



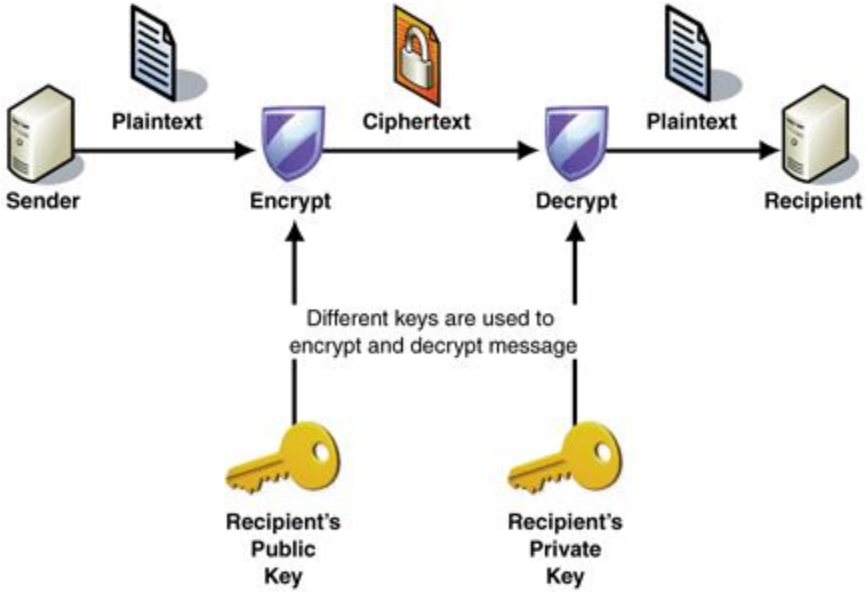
Hashing and Salting Passwords

- Hashing = one-way transformation
- Use for storing passwords (not reversible)
- Salting: add random data to prevent rainbow table attacks



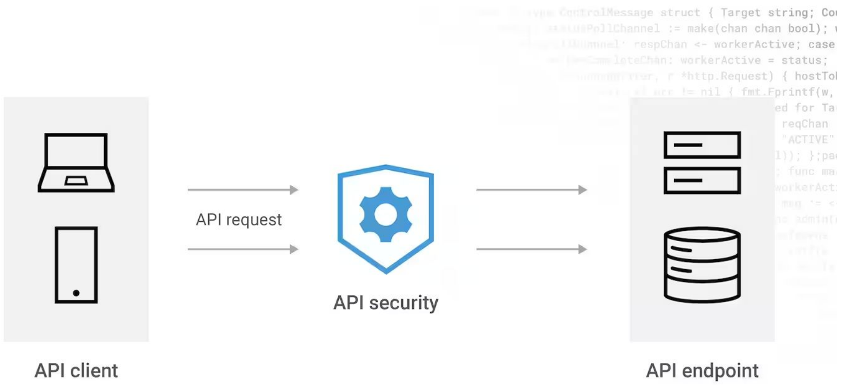
Public Key Infrastructure (PKI)

- PKI = system for managing digital certificates and keys
- Role of Certificate Authorities (CAs)
- Digital signatures & certificate chains



Secure API Communication

- Use HTTPS for all API traffic
- Auth tokens (JWT, OAuth)
- Rate limiting, IP whitelisting, mutual TLS for sensitive APIs



Interview Questions

- What is the difference between hashing and encryption?
- Why is asymmetric encryption slower than symmetric?
- How does PKI build trust online?
- What is the concept behind securing Data at rest and Data in Motion?

Summary & Best Practices

- Encrypt data both at rest and in transit
- Use hashing + salting for passwords
- Leverage TLS, PKI, HTTPS for secure communication
- Harden APIs with secure design patterns
- What’s next:
  - Network & Infrastructure Security

Network & Infrastructure Security

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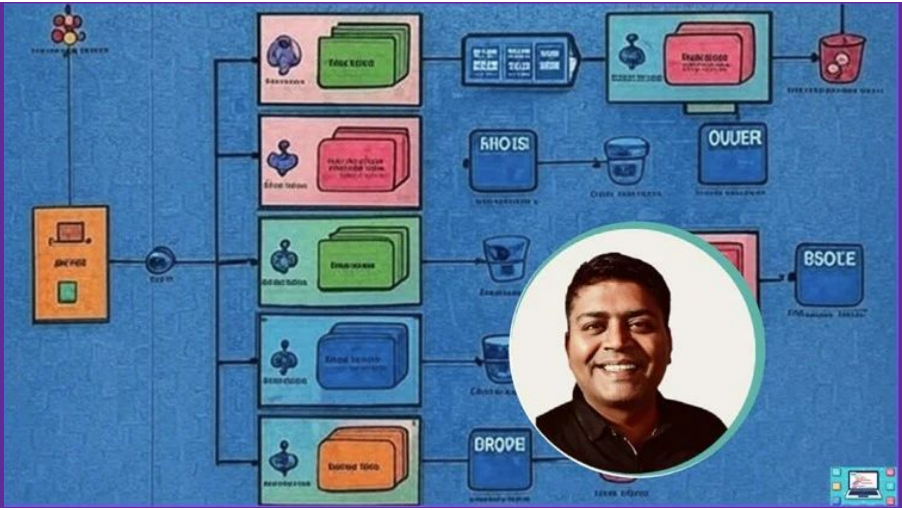
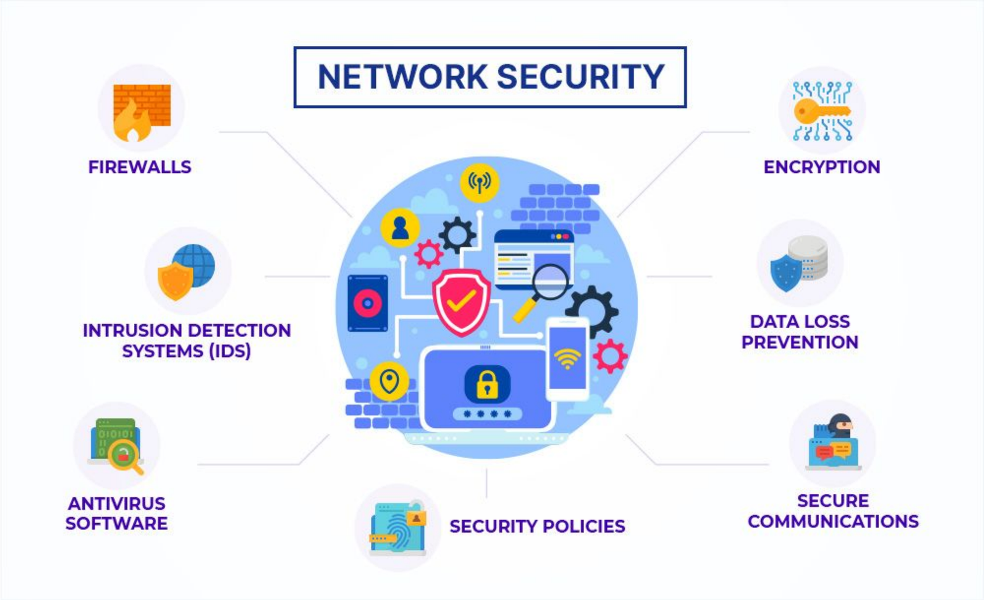


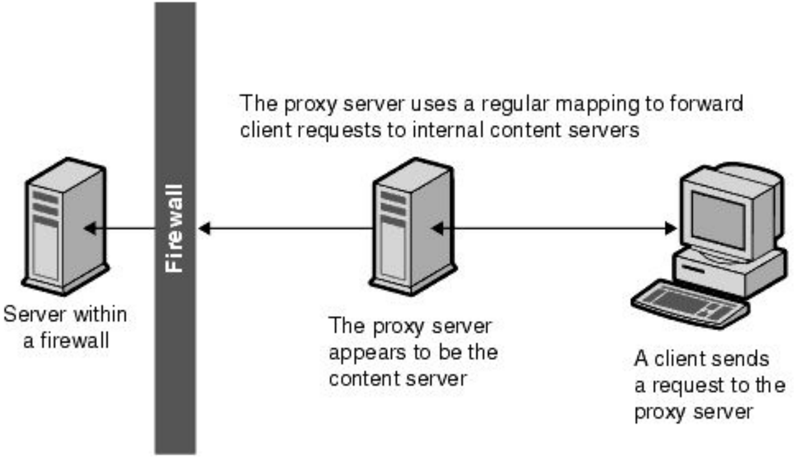
Why Network Security Matters

- External threats (DDoS, intrusion, IP spoofing)
- Internal risks (misconfiguration, lateral movement)
- Increasing cloud-native adoption = more exposure
- Reliability & user trust



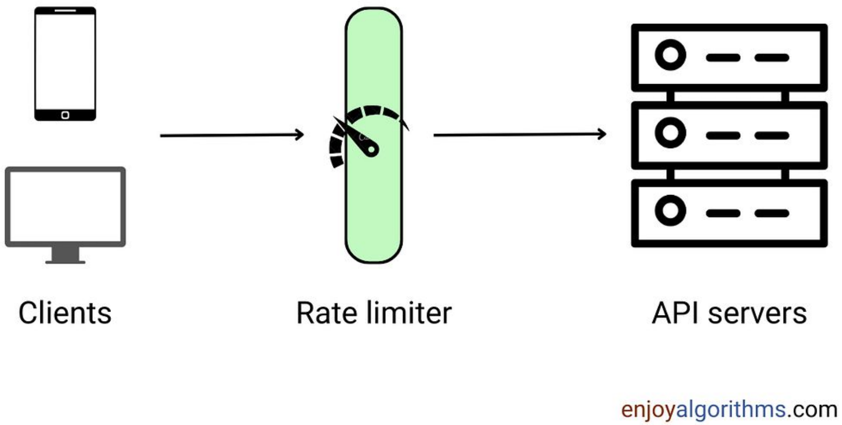
Firewalls and Reverse Proxies

- Firewalls: filter traffic based on IP, port, protocol
- Types: Network-based, Host-based, Cloud firewalls
- Reverse Proxies: route, mask backend identity, add security
- Examples: NGINX, AWS ALB



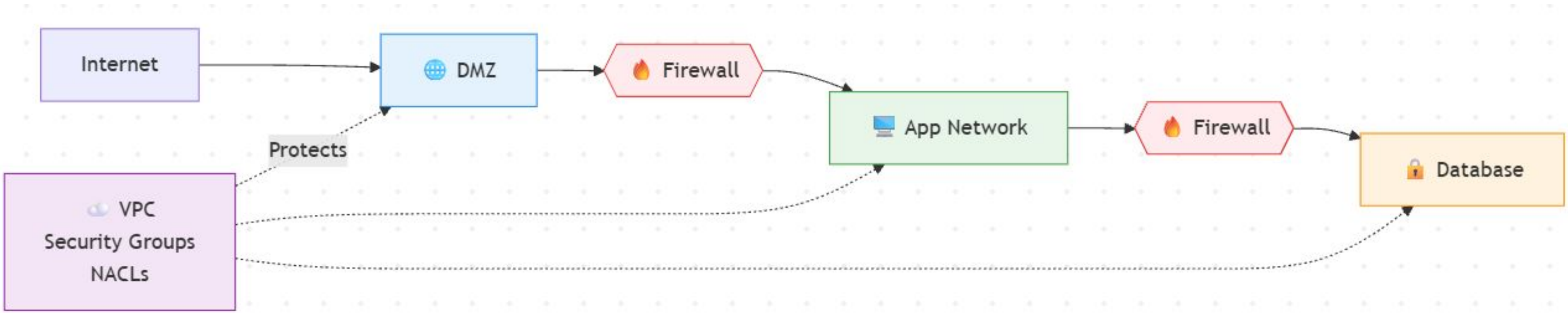
Rate Limiting, Throttling, IP Filtering

- Rate limiting: per-user, per-IP request caps
- Throttling: graceful degradation under load
- IP Filtering: allow/block lists
- Helps protect APIs & backend systems from abuse



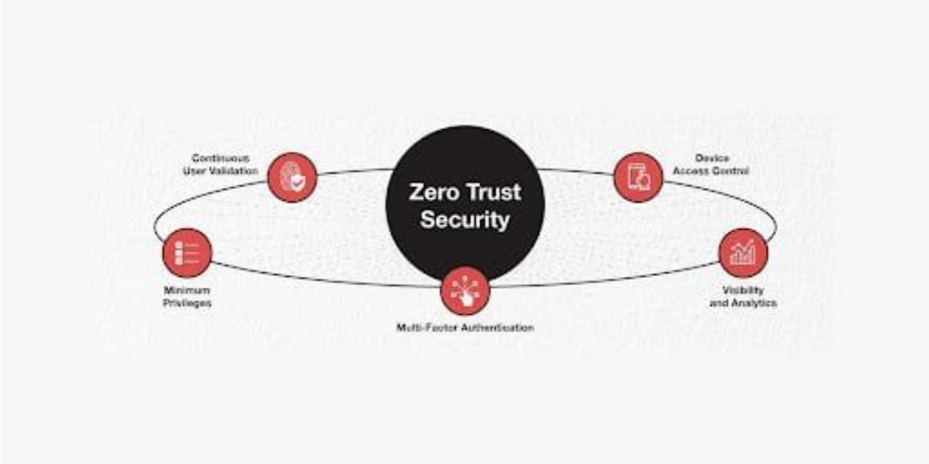
Network Segmentation & Isolation

- Split network into zones: DMZ, internal, DB, etc.
- Limit lateral movement
- Use firewalls, subnets, private VLANs
- Cloud: use VPCs, security groups, NACLs



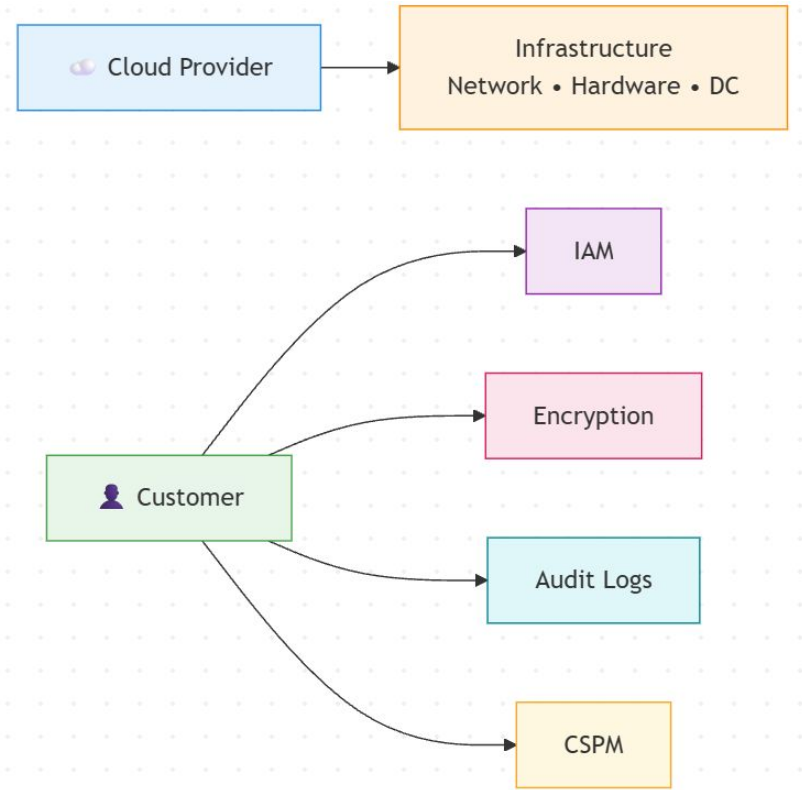
Zero Trust Security Model

- “Never trust, always verify”
- Auth every request, even inside the network
- Microservices: mutual TLS, strict access control
- Applies to cloud, hybrid, and on-prem setups



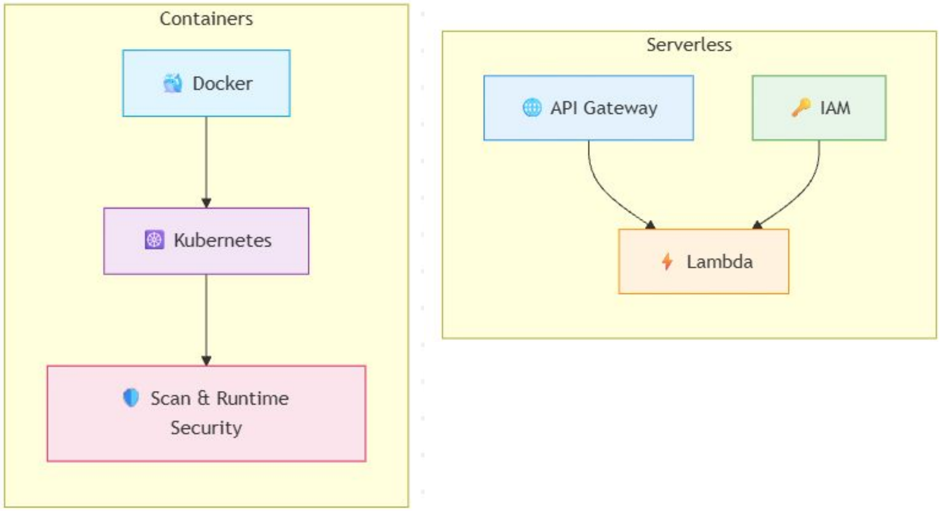
Securing Cloud Environments

- Shared responsibility model
- Key aspects:
  - IAM
  - Encryption (EBS, S3, RDS)
  - Audit logging (CloudTrail, Stackdriver)
- CSPM (Cloud Security Posture Management) tools



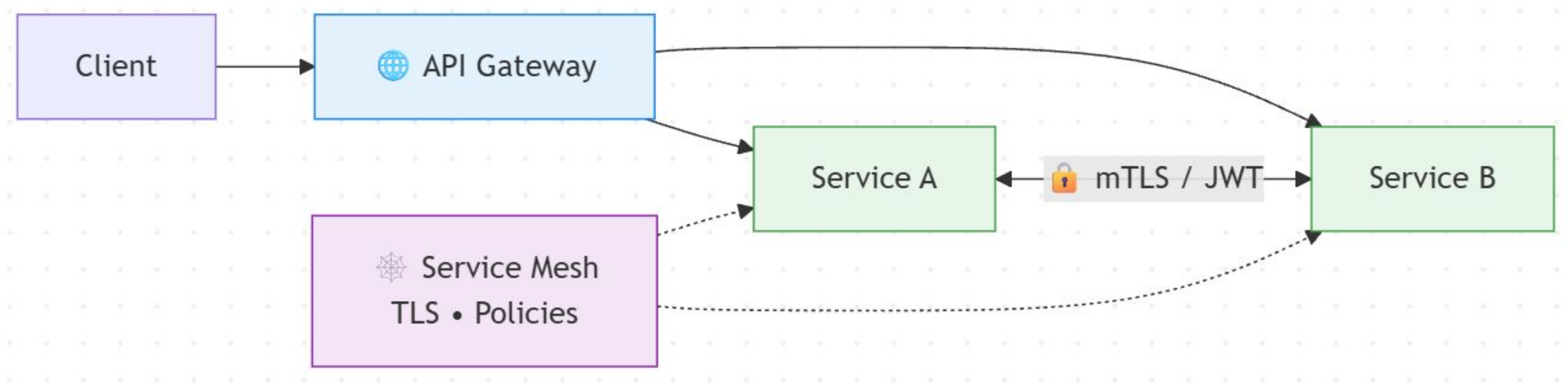
Securing Serverless & Containerized Workloads

- Serverless: control IAM roles, timeouts, API gateway access
- Containers: image scanning, runtime hardening, least privilege
- Tools: AWS Lambda + API Gateway, Docker + Kubernetes security tools (OPA, Falco)



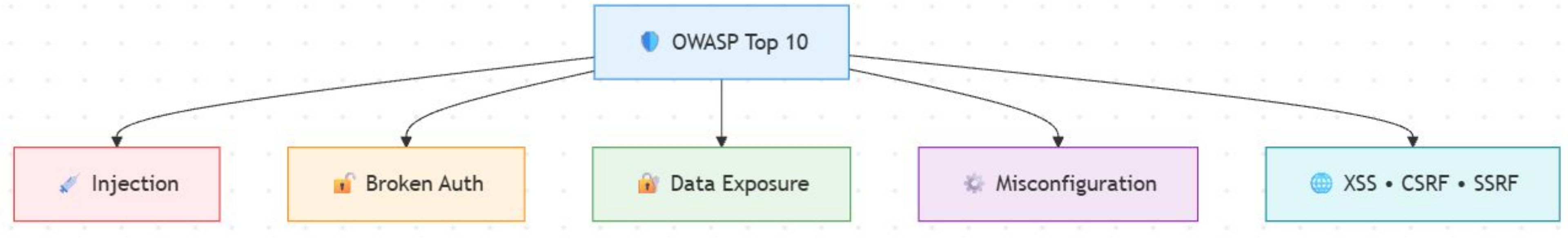
Security in Microservices

- Service-to-service auth (JWT, mTLS)
- API Gateway security: validation, auth, rate limits
- Service mesh: fine-grained control, TLS, policies (Istio, Linkerd)



Common Vulnerabilities (OWASP Top 10)

- A quick glance at top risks:
  - Injection
  - Broken Auth
  - Sensitive Data Exposure
  - Security Misconfig
  - XSS, CSRF, SSRF, etc.
- Relevance in web apps & microservices
- Focus on awareness + mitigation



Interview Questions - Network Security

- What is the difference between a firewall and a reverse proxy?
- How does rate limiting protect your backend services?
- Explain the Zero Trust security model and its importance.
- How would you secure a containerized workload in Kubernetes?
- What are some common OWASP Top 10 vulnerabilities and how do you mitigate them?
- How does a service mesh help enforce security in microservices?
- What’s the role of IAM in cloud security?

Summary & Best Practices

- Use firewalls and segment your network
- Apply rate limiting, IP filters, reverse proxies
- Embrace Zero Trust and encrypt everywhere
- Secure APIs, services, containers, and functions
- Stay alert to OWASP Top 10
- What’s next:
  - Summary and Recap: Designing Secure Distributed Systems

Section Summary: Security in System Design

- Security in Distributed Systems
  - CIA Triad: Confidentiality, Integrity, Availability
  - Common Threats: DDoS, MITM, Injection, Spoofing
- Authentication & Authorization
  - Auth Methods: OAuth2, OpenID, JWT
  - Access Control: RBAC, ABAC, SSO
- Data Protection & Secure Communication
  - Encryption: At rest & in transit
  - TLS/SSL, HTTPS, Hashing, Salting
- Network & Infrastructure Security
  - Firewalls, Reverse Proxies, Rate Limiting
  - Network Segmentation, Zero Trust
  - Cloud & Microservices Security: IAM, Encryption, API Gateway
- What’s next:
  - The System Design Blueprint